discussion session : Mar. 25, 2024

textbook : 11.1
11.2
11.3

ex 1: Find limit of each sequence :

a) $a_n = \dfrac{3n^2+16n+\frac{1}{n}}{(4n-1)(\frac{1}{2}n+1)}$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{3n^2+16n+\frac{1}{n}}{(4n-1)(\frac{1}{2}n+1)} = \lim_{n\to\infty} \frac{3n^2+16n+\frac{1}{n}}{2n^2-\frac{1}{2}n+4n-1} = \frac{3}{2}$$

at large n, $\approx \dfrac{3n^2}{2n^2}$

b) $a_n = \dfrac{6^{3n}}{4^n}$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{6^{3n}}{4^n} = \lim_{n\to\infty} \left(\frac{6^3}{4}\right)^n \text{ diverges}$$

$\frac{6^3}{4} > 1$

c) $a_n = \dfrac{(-1)^{n+1}\, 6\ln(n+4)}{6(n+4)}$

Use th'm: if $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$.

$$\lim_{n\to\infty} |a_n| = \lim_{n\to\infty} \frac{6\ln(n+4)}{6(n+4)} = \lim_{n\to\infty} \frac{\frac{6}{n+4}}{6} = \lim_{n\to\infty} \frac{1}{n+4} = 0.$$

L'Hopital's

So, $\lim_{n\to\infty} a_n = 0$.

d) $a_n = 2 + \dfrac{4n^2}{2^n}$.

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}\left(2+\frac{4n^2}{2^n}\right) = \lim_{n\to\infty}(2) + \lim_{n\to\infty}\left(\frac{4n^2}{2^n}\right) = 2 + \lim_{n\to\infty}\left(\frac{4n^2}{2^n}\right)$$

$$= 2 + \lim_{n\to\infty}\left(\frac{8\cancel{n}}{\cancel{n}\cdot 2^{n-1}}\right)$$

L'H.

$$= 2$$

ex 2: Determine if the sequence is increasing, decreasing, or not monotonic.

a) $a_n = \frac{5-n}{6+n}$

$$a_{n+1} = \frac{5-(n+1)}{6+(n+1)} = \frac{4-n}{7+n}$$

$$a_{n+1} - a_n = \frac{4-n}{7+n} - \frac{5-n}{6+n} = \frac{(4-n)(6+n) - (5-n)(7+n)}{(7+n)(6+n)}$$

$$= \frac{24 - 2n - n^2 - (35 - 2n - n^2)}{(7+n)(6+n)}$$

$$= \frac{-11}{(7+n)(6+n)} < 0 \quad \text{for all } n = 1, 2, 3, \ldots$$

So, $a_{n+1} < a_n$, meaning the sequence decreases.

b) $a_n = n - \frac{1}{n}$

$$a_{n+1} = n+1 - \frac{1}{n+1}$$

$$a_{n+1} - a_n = n+1 - \frac{1}{n+1} - n + \frac{1}{n} = 1 - \frac{1}{n+1} + \frac{1}{n} = 1 + \frac{(-n)+(n+1)}{(n+1)n} = 1 + \frac{1}{n(n+1)} > 0$$

So, $a_{n+1} > a_n$, meaning the sequence increases.

c) $a_n = \frac{1}{2n+3}$

$$a_{n+1} = \frac{1}{2(n+1)+3} = \frac{1}{2n+5}$$

$$\frac{a_{n+1}}{a_n} = \frac{\frac{1}{2n+5}}{\frac{1}{2n+3}} = \frac{2n+3}{2n+5} < 1 \quad \text{for } n = 1, 2, 3, \ldots$$

So then $a_{n+1} < a_n$, meaning the sequence decreases.

Note about bounded sequences:

A sequence is bounded if it is bounded above and below. That means that there exists a number M and a number N such that

$$M \le a_n \le N \quad \text{for all } n \ge 1.$$

So, the sequence in 2(c) is bounded:

$a_n = \frac{1}{2n+3}$ decreases, so the max. is at $n=1$: $a_1 = \frac{1}{5}$. (upper bound)

Since $\lim_{n\to\infty} a_n = 0$, it is bounded below, too. $\frac{1}{5} \ge a_n \ge 0$ for all $n \ge 1$.

ex 3: Integral test

a) $\sum_{n=1}^{\infty} \frac{1}{(n+5)^2}$

since $f(x)$ is a square

since $\frac{1}{(x+1+5)^2} \le \frac{1}{(x+5)^2}$

$f(x) = \frac{1}{(x+5)^2}$ is a positive, decreasing, continuous function for $x \in [1, \infty)$

(Notice that the only discontinuity would be at $x = -5$, which is not in $[1, \infty)$.)

So, use integral test to test for convergence:

$$\int_1^{\infty} \frac{1}{(x+5)^2}\,dx = \int_6^{\infty} \frac{1}{u^2}\,du = \lim_{t\to\infty} \int_6^t \frac{1}{u^2}\,du = \lim_{t\to\infty} \left[\frac{u^{-1}}{-1}\right]_6^t$$

$$\begin{bmatrix} u = x+5 \\ du = dx \end{bmatrix}$$

$$= \lim_{t\to\infty} \left[-\frac{1}{t} + \frac{1}{6}\right] = \frac{1}{6}$$

So, since this integral converges, $\sum_{n=1}^{\infty} \frac{1}{(n+5)^2}$ converges.

b) $\sum_{n=1}^{\infty} \frac{1}{n}$ ← Notice that this is a p-series where $p = 1$.

(By the test for divergence, we don't know yet if this is a convergent or divergent series, since $\lim_{n\to\infty} a_n = 0$.)

$f(x) = \frac{1}{x}$ is a positive, decreasing, continuous function for $x \in [1, \infty)$

since x is positive

since $\frac{1}{x} > \frac{1}{x+1}$

only discontinuity would be at $x = 0$

So, use the integral test:

$$\int_1^{\infty} \frac{1}{x}\,dx = \lim_{t\to\infty} \int_1^t \frac{1}{x}\,dx = \lim_{t\to\infty} \ln x \Big|_1^t = \lim_{t\to\infty} (\ln t - \cancel{\ln 1}^{\,0})$$

diverges

So, $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

c) $\sum_{n=1}^{\infty} \frac{\ln n}{n}$

$f(x) = \frac{\ln x}{x}$ is positive, decreasing, and continuous on $x \in [1, \infty)$,

so $$\int_1^{\infty} \frac{\ln x}{x}\,dx = \int_0^{\infty} u\,du = \lim_{t\to\infty} \int_0^t u\,du = \lim_{t\to\infty} \frac{u^2}{2}\Big|_0^t = \lim_{t\to\infty} \left(\frac{t^2}{2}\right)$$

diverges

$$\begin{bmatrix} u = \ln x \\ du = \frac{1}{x}\,dx \end{bmatrix}$$

So, $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ diverges.

ex 4: Determine if the series converges or diverges. If it converges, find the limit.

a) $\sum_{n=1}^{\infty} 5^{n-1}$ is a geometric series $\sum_{n=1}^{\infty} ar^{n-1}$ with $|r| = |5| \geq 1$.

So, this is a divergent series.

b) $\sum_{n=1}^{\infty} \left(\frac{1}{5}\right)^{n-1}$ is a geometric series with $|r| = \left|\frac{1}{5}\right| < 1$, so this is a convergent series. Also, $\sum_{n=1}^{\infty} = \frac{1}{1-\left(\frac{1}{5}\right)} = \frac{1}{\frac{4}{5}} = \frac{5}{4}$.

c) $\sum_{n=1}^{\infty} \sin n$. Use test for divergence: $\lim_{n \to \infty} \sin n$ diverges, so the series diverges.

d) $\sum_{n=1}^{\infty} \frac{n^2}{5n^2+1}$. Use test for divergence: $\lim_{n \to \infty} \frac{n^2}{5n^2+1} = \frac{1}{5} \neq 0$, so the series diverges.

e) $\sum_{n=1}^{\infty} \left(\frac{1}{5}\right)^{2n} = \sum_{n=1}^{\infty} \left(\frac{1}{25}\right)^{n} = \sum_{n=1}^{\infty} \left(\frac{1}{25}\right)\left(\frac{1}{25}\right)^{n-1} = \frac{1/25}{1-1/25} = \frac{1}{24}$ is a convergent geometric series. ($|r| = 1/25$)